

Hakata old town, Fukuoka

Then-and-now aerial photographs, how the ground was made, and the disasters people chose to remember — compiled automatically from public Japanese government data.



博多（福岡市博多区）

Study area	2.2 km × 1.4 km around 33.5930° N, 130.4110° E
Oldest aerial photograph	1945–1950
Aerial series in this brief	1945–1950, 1961–1969, 1974–1978, 1987–1990, 2020
Main natural landform	Sandbars and dunes (72%)
Former river channels	none mapped
Filled, reclaimed or drained land	16%
Former sea, river or pond	under 0.5%
Meiji-era paddies, marsh and flats	rice paddy 25%, sand and gravel 1.6%, tidal flat or beach 1.4%
Disaster memorials within 3 km	0

What this brief is. A reading aid for comparing the same ground across decades. It shows what the photographs and GSI's land surveys record; it does not describe current conditions, access or opening hours.

What it is not. Not a hazard assessment or a property survey. Landform classes describe typical tendencies of the ground; for present-day risk, read the official hazard map of the municipality.

Edition 2026-09-27. Aerial photographs, landform and Meiji-era lowland data: Geospatial Information Authority of Japan (GSI), processed by Japan Time Atlas. Memorial data: GSI Natural Disaster Memorials, summarised and translated by Japan Time Atlas.

The same ground, 1945–1950 and 2020



1945–1950 aerial photograph (US military photography) — GSI Tiles *ort_USA10*, cropped and annotated by Japan Time Atlas. North is up; the frame is identical in both images.



2020 annual orthophoto — GSI Tiles *nendophoto2020*, cropped and annotated by Japan Time Atlas. Capture dates within a mosaic can differ from place to place.

How to compare. Anchor on features that rarely move — river banks, rail lines, shrine and temple grounds, main roads — then look for what changed around them. Differences in tone come from film, camera and season, not from the land itself.

The decades in between

Every further GSI series that covers the same frame, oldest first. Each series is a compilation: photographs within one series were taken on different dates, which is why seams and changes of tone can appear inside a frame.



1961–1969 · GSI Tiles *ort_old10*, processed

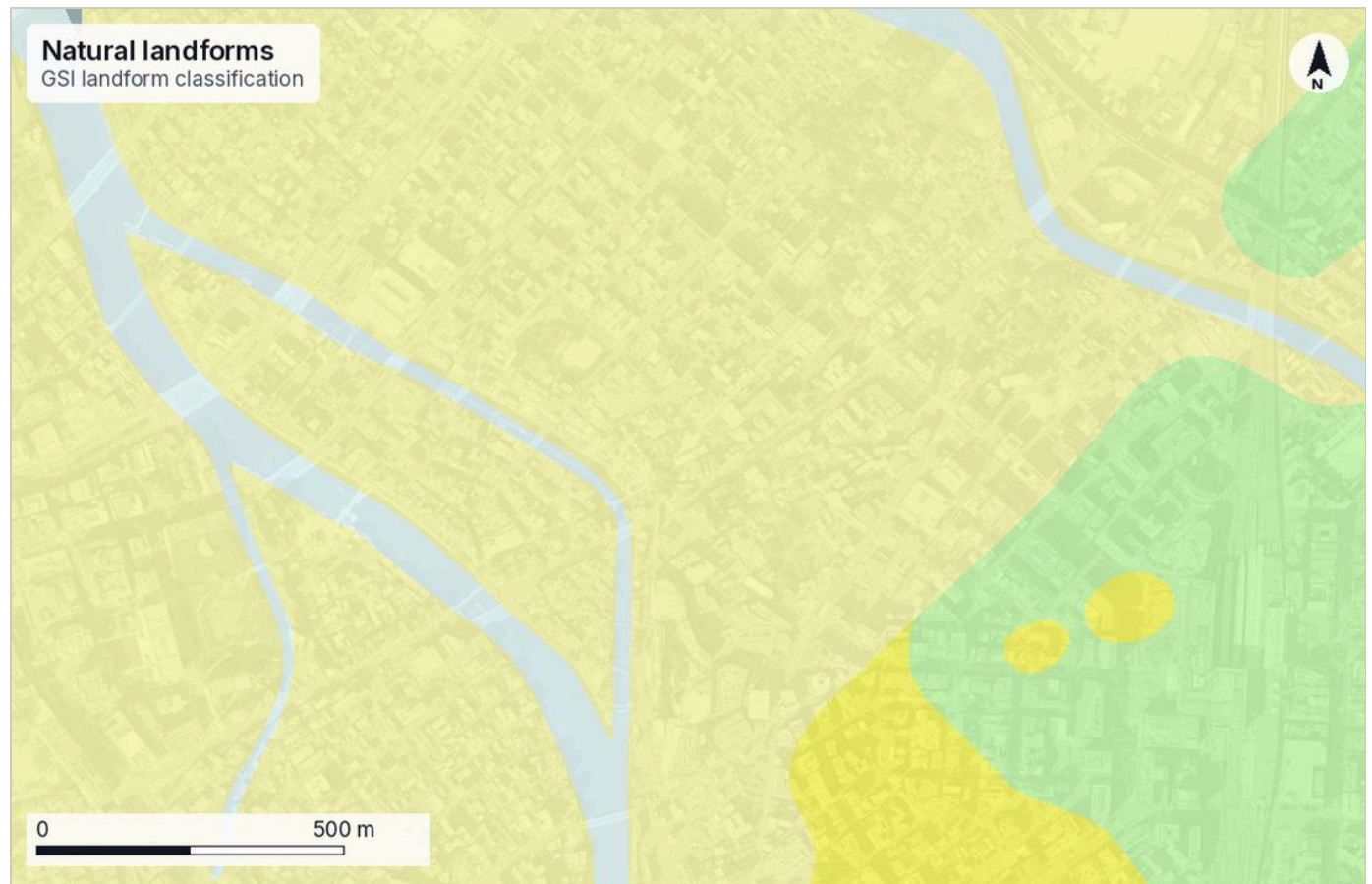


1974–1978 · GSI Tiles *gazo1*, processed



1987–1990 · GSI Tiles *gazo4*, processed

Rivers, shorelines and terraces



Natural landforms as classified by GSI, over a recent aerial photograph in gray. Dark blue outlines mark former river channels.



What the main landforms mean

Sandbars and dunes (72%). Sandy or gravelly ground higher than its surroundings along present or former coasts, lakeshores and riverbanks, built by waves or wind. Usually escapes ordinary floods. Liquefaction tendency is somewhat strong on sandbars, somewhat weak on dunes and very strong at dune margins.

Floodplain and coastal plain (16%). Low, flat ground of little relief, built from sand and mud left by floods, or former seabed that emerged. River flooding. Ground is softer nearer the coast and shakes somewhat more; liquefaction tendency is strong on coastal plains and somewhat strong on floodplains. Storm surge near the coast.

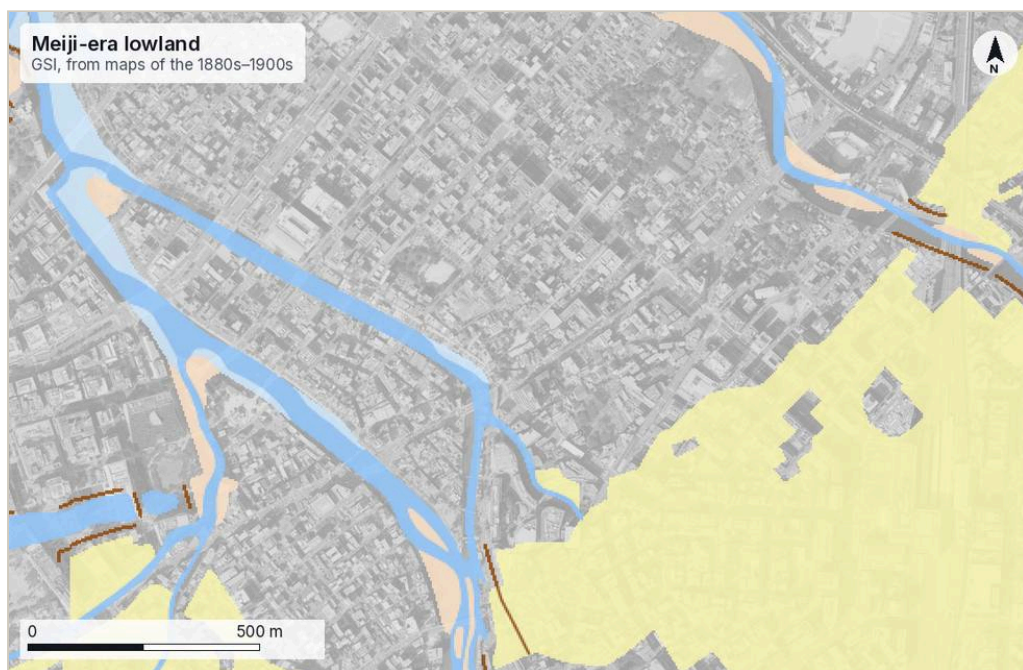
Natural levee (4.6%). Long, low ridges 0.5 m to a few metres above their surroundings along present or former rivers, built from sediment dropped by floods. Usually above ordinary floods but can be covered in major ones; strong liquefaction tendency at the edges.

Descriptions translated and condensed from GSI's landform-classification legend. They describe tendencies of each landform type, not the condition of any individual plot.

Fill, reclamation and the fields beneath the town



Hatching: man-made ground as classified by GSI. Teal outlines: former sea, river or pond. Dark blue outlines: former river channels.



GSI “Meiji-era lowland” data, traced from topographic maps of the 1880s–1900s; GSI warns that positions can be off by a considerable distance, especially in the Kanto and Kinki regions. Only low, wet or open ground was traced: towns, dry fields and forest of the period are left blank.

Man-made ground

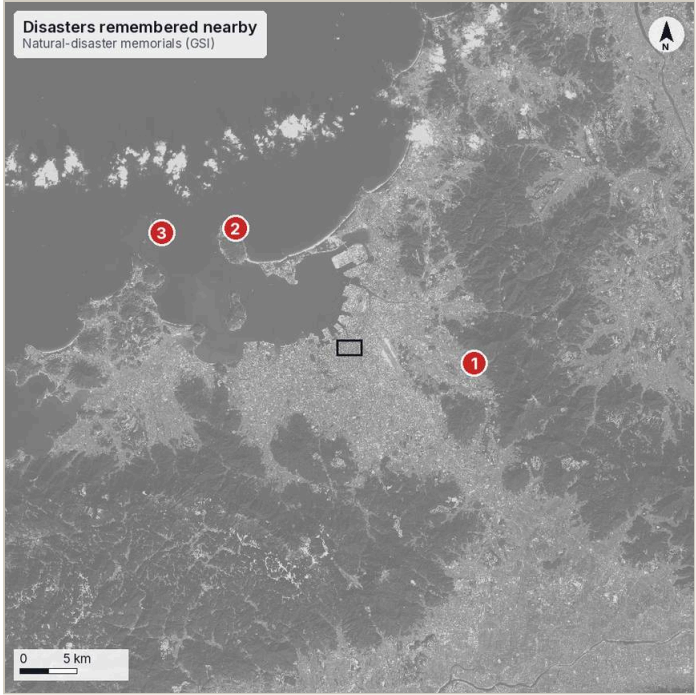
 Filled and reclaimed land 16%

Meiji-era lowland

 Rice paddy 25%
 River, pond or sea 5.1%
 Sand and gravel 1.6%
 Tidal flat or beach 1.4%

Filled and reclaimed land (16%). Ground raised with fill, land made by dumping soil into the sea, rivers or ponds, or valleys filled in. May flood if not raised enough; on hills, rain or earthquakes can make the fill fail; in lowlands the liquefaction tendency is very strong, especially on filled sea, lake or river beds.

Memorials to natural disasters nearby



Numbered memorials; the black rectangle is the study area of this brief. Frame 60 km across.

Fewer than three memorials stand within 3 km of the study area, so the nearest ones within 30 km are listed.

	Memorial	Disaster	Erected	Distance
1	Monument to the recovery from the 1973 flood in Satani 水害復興碑	Torrential rain (31 Jul 1973) · flood / landslide or debris flow	1976	11.0 km
2	Monument to the recovery from the 2005 Fukuoka earthquake (Katsuma) 福岡県西方沖地震復興記念碑	Earthquake west off Fukuoka Prefecture (20 Mar 2005) · earthquake	2006	14.4 km
3	Monument to Genkai Island's recovery from the 2005 Fukuoka earthquake 福岡県西方沖地震復興記念碑	Earthquake west off Fukuoka Prefecture (20 Mar 2005) · earthquake	2008	19.2 km

What the memorials record

- 1. Monument to the recovery from the 1973 flood in Satani.** Before dawn on 31 July 1973 torrential rain dropped 239 mm on Sue in just two hours. In the Satani district, debris flows and other damage killed 2 people, destroyed 21 houses and half-destroyed 6, flooded 31, and washed away or buried 23.8 hectares of farmland.
- 2. Monument to the recovery from the 2005 Fukuoka earthquake (Katsuma).** On 20 March 2005 a magnitude 7.0 earthquake centred in the Genkai Sea off Fukuoka City, with a seismic intensity of 6 lower on the Japanese scale, closed roads with slope collapses and damaged houses.
- 3. Monument to Genkai Island's recovery from the 2005 Fukuoka earthquake.** On 20 March 2005 a magnitude 7.0 earthquake centred in the Genkai Sea off Fukuoka City, with a seismic intensity of 6 lower on the Japanese scale, devastated Genkai Island: 19 people were injured and 214 homes damaged, 107 of them completely destroyed.

Source: GSI Natural Disaster Memorials (自然災害伝承碑). English names and notes are Japan Time Atlas translations and summaries of GSI's Japanese descriptions; the memorials' own inscriptions may say more. "Erected" is the year GSI records for the monument.

Where every element comes from

- **Aerial photographs** (pp. 2–3) — Geospatial Information Authority of Japan (GSI), GSI Tiles: series *ort_USA10* (1945–1950, US military photography), *ort_old10* (1961–1969), *gazo1* (1974–1978), *gazo4* (1987–1990) and *nendophoto2020*. Cropped, resized and annotated by Japan Time Atlas. List of GSI Tiles: <https://maps.gsi.go.jp/development/ichiran.html>
- **Landform classification** (pp. 4–5) — GSI landform-classification vector tiles (natural and artificial landforms), compiled by GSI from its Land Condition Maps, Flood Control Terrain Classification Maps and vulnerable-landform surveys. Rendered, measured and translated into English by Japan Time Atlas. https://github.com/gsi-cyberjapan/experimental_landformclassification
- **Meiji-era lowland** (p. 5) — GSI “明治期の低湿地” tiles and legend (https://maps.gsi.go.jp/legend/lw_legend.pdf), measured and translated by Japan Time Atlas.
- **Natural disaster memorials** (p. 6) — GSI Natural Disaster Memorials (自然災害伝承碑), registered by municipalities, <https://www.gsi.go.jp/bousaichiri/denshouhi.html>; processed by Japan Time Atlas (selection by distance, English translation and summary). Memorial photographs are not reproduced.
- GSI content is used under the GSI Content Terms of Use, which follow the Public Data License v1.0 (PDL1.0, compatible with CC BY 4.0). Data retrieved 2026-09-27.
- **Outline of Japan** — Data of Japan / JapanPrefGeoJson (public domain), simplified.

Method

The study area is a fixed rectangle of 2.2 × 1.4 km. Every GSI photo series that covers at least 85% of it is included; the newest annual orthophoto covering at least 90% of it is used as “now”, with any gap filled from GSI's latest seamless mosaic. Landform shares are areas of the rectangle measured on GSI's polygons at zoom level 14, each class merged before measuring. Meiji-era shares count map pixels by legend color. Memorials are selected by straight-line distance from the center of the study area.

Limits

Historical series are mosaics of photographs taken on different dates; GSI publishes the flight date of each photograph. Landform data record conditions at the time of survey. Nothing in this brief shows present-day access, safety or opening hours.

Area Dossier. The same district with a 4–8 page English history, written from Japanese library sources and cited page by page.

Deep Research. A single hamlet or address, researched in libraries and archives.

Compiled by Japan Time Atlas; not produced, reviewed or endorsed by GSI, MLIT or any Japanese government agency. Not an official hazard map and not for statutory real-estate disclosure. Generated 2026-09-27 by an automated pipeline (research/lab-07/scripts/basic.py).